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Density structure of Kīlauea volcano: implications for magma storage and transport

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SUMMARY

A Bayesian linear regression to determine the bias in the Nafe–Drake relationship between compressional velocity and density provides an improved model for the density structure of Kīlauea volcano, Hawai'i. In previous work, we combined the results of seismic tomography with the Nafe–Drake relationship between compressional velocity and density to explain the large values of gravity disturbances overlying the summits and rift zones of the island's volcanoes. These results were used to determine mechanisms for gravitational instability of the island flanks. Here, we use laboratory measurements of the relationship of velocity and density for a wide range of Hawai'i island rocks as a prior in a Bayesian regression, with seismic tomography, to refine the 3-D density structure for Kīlauea volcano. This refined structure shows dense bodies (3220 kg m^{-3}) between 5 and 8 km below sea level that underly regions of magma storage, found from geodetic and geophysical studies, beneath the summit and East Rift Zone of Kīlauea volcano. Above these bodies, density isosurfaces surround and cradle sources of pressure change determined from geodetic models, both at the summit and along the East Rift Zone. Continued subsidence of the summit following the 2018 eruption is aligned with a bowl-shaped density structure, formed primarily by density isosurfaces between 2800 and 2900 kg m^{-3} at 4–6 km depth. These surfaces underly the ∼3 km depth at which dyke injection initiates, are largely aseismic, and from their density values are inferred to contain high concentrations of olivine. Taken together, these density structures are consistent with an olivine-rich mush with variable porosity that increases in density with depth and provides a mechanism to form olivine cumulates both at the summit and along the rift zones. This structural framework for Kīlauea volcano is consistent with melt and mush transport occurring over a large range of depths to accommodate the growth and spreading of the volcano.

Key words: Gravity anomalies and Earth Structure; Magma migration and fragmentation; Physics and chemistry of magma bodies; Volcano seismology.

INTRODUCTION

The surface structure of Kīlauea volcano is evident from surface faulting and topography (Denlinger & Okubo 1995, Fig. 1), from observations of dykes exposed in the summit caldera (Casadevall & Dzurisin 1987), and from dykes along the East Rift Zone (ERZ) (Montgomery-Brown *et al.* 2010). The internal structure of Kīlauea volcano is much less evident, and is interpreted from geological and geophysical measurements (Zucca & Hill 1980; Ryan 1988; Hildenbrand *et al.* 1993; Kauahikaua *et al.* 2000; Brooks *et al.* 2006; Montgomery-Brown *et al.* 2009, 2010, 2015; Flinders *et al.* 2010, 2013; Anderson *et al.* 2015; Ge *et al.* 2019; Segall *et al.* 2019; Shelly & Thelen 2019; Lerner *et al.* 2024). The summit and ERZ are connected by nearly continuous gravity and magnetic anomalies (Hildenbrand *et al.* 1993; Kauahikaua *et al.* 2000). The magnetic anomaly of the summit and ERZ is related to dyke structure (Hildenbrand *et al.* 1993), and the axis formed by dykes overlies a deep zone of magma storage along the ERZ, as shown from the subsidence during the 2018 eruption (Fig. 2). Gravity anomalies at the summit and along the rift zone are associated with olivine accumulation within a magma system at depth (Clague & Denlinger 1994; Kauahikaua *et al.* 2000; Flinders *et al.* 2010). Previously, Denlinger & Flinders (2022) showed that the Nafe–Drake velocity–density relationship, when combined with the seismic tomography results of Lin *et al.* (2014), produced a density structure that approximated the observed gravity disturbance field on the island. This

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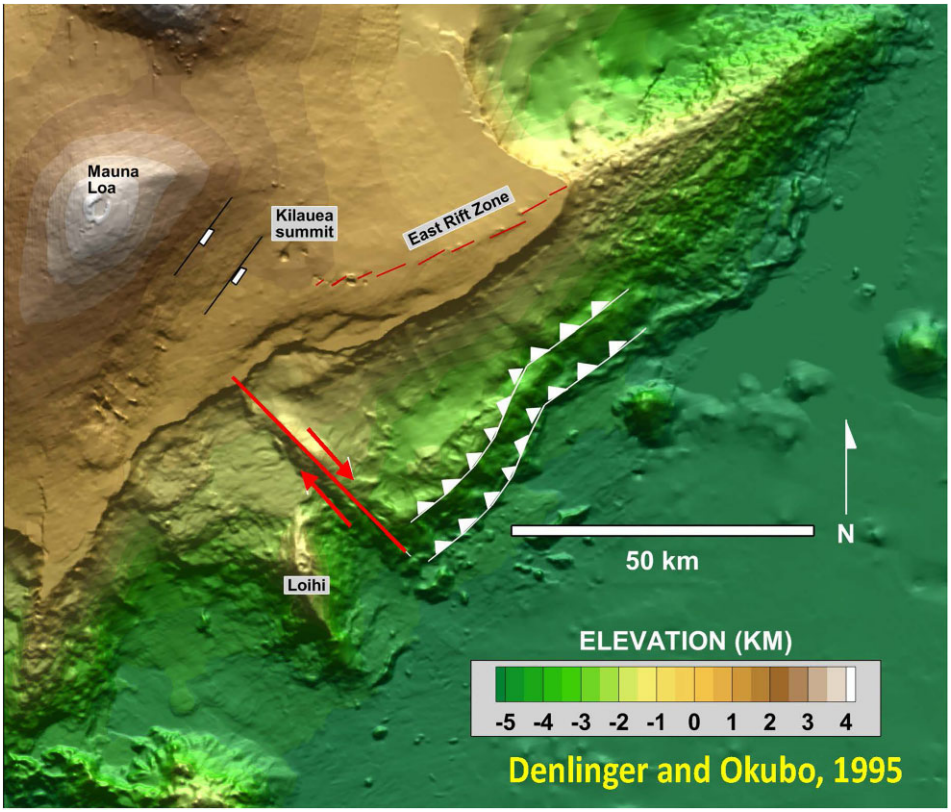

Figure 1. Surface structure of the southeast portion of the island of Hawai'I, adapted from Denlinger & Okubo (1995). The ERZ of Kīlauea volcano, shown as a series of red dashes to highlight the wrench fault structure, heads southeast from Kīlauea summit before turning east, then northeastward and continuing eastward offshore as the prominent submarine ridge shown here. The motion of the south flank of Kīlauea is about S20°E, and subparallel to the S50°E strike of the right lateral slip fault that forms its western margin with Kama'ehuakanaloa (formerly Lō'ihi). Separation of Kīlauea from Mauna Loa causes wrench faulting on adjacent Mauna Loa, consisting of both extension (shown) and right lateral strike-slip. The large prominent fault scarps near the coast south of the summit are the Hilina Pali fault system, and we constrain the low-density roots of this system in this report.

correspondence was then used to determine the relative importance of gravitational loading and magma pressure on the instability of the island flanks.

Though the modelled gravity field was previously found to be within 90 per cent of the observed field for elevations below 1 km (Denlinger & Flinders 2022), the densities estimated using the Nafe–Drake relationship are too low. High, mantle-like, compressional velocities of more than 7.2 km s^{-1}, beneath the summits of Kīlauea and Mauna Loa volcanoes were associated with a density of only 3027 kg m^{-3}. Yet many field data indicated that higher densities are common: picrites have been emplaced in the summit (Casadevall & Dzurisin 1987) and also comprised some lava erupted from Kīlauea Iki in 1959 (Helz 1987). These observations, combined with high seismic velocity structure beneath the summit (Lin *et al.* 2014), high gravity anomalies over the summit (Kauahikaua *et al.* 2000), and petrologic evidence from erupted lava support the existence of dense olivine cumulates within the Kīlauea magma system (Clague & Denlinger 1994). The low densities that Denlinger & Flinders (2022) used were obtained using the global Nafe–Drake velocity–density relationship with a zero bias. Yet in fitting their lab measurements of velocity and density for Hawaiian rocks, using the same Nafe–Drake relationship, Manghnani & Woollard (2013) found it necessary to add a constant bias of 220 kg m^{-3}.

We test the efficacy of a bias in the Nafe–Drake relationship for all Hawaiian rocks using field measurements of gravity disturbances combined with the results of traveltime seismic tomography, as we did previously. However, now we use laboratory measurements between compressional velocity and density as a prior constraint, and determine the Bayesian posterior probability of a positive density bias. This approach produces a more robust assessment of the bias, since the lab-determined bias of 220 kg m^{-3} presented by Manghnani & Woollard (2013, or any laboratory measurement) could be subject to a scaling error resulting from extrapolating density of small, intact, lab samples, to that of the entire island. Our results sample much larger volumes (by 5 orders of magnitude) and allow us to test the conversion of the seismic tomography structure for Kīlauea volcano into a detailed density structure. The resulting provocative density structure for the summit and ERZ of the volcano forms an insightful and useful structural framework for continued seismic, geodetic and petrologic studies.

METHOD

To accommodate a constant bias in the relationship between compressional velocity and density, we generalize the polynomial expansion describing the Nafe–Drake relationship. As given by Brocher (2005), the relationship for density ρ at any subsurface location is obtained from the compressional velocity v_p at that location by the polynomial series

$$\rho(\mathbf{v}) = \sum_{1}^{5} w_i \phi_i(\mathbf{v}), \tag{1}$$

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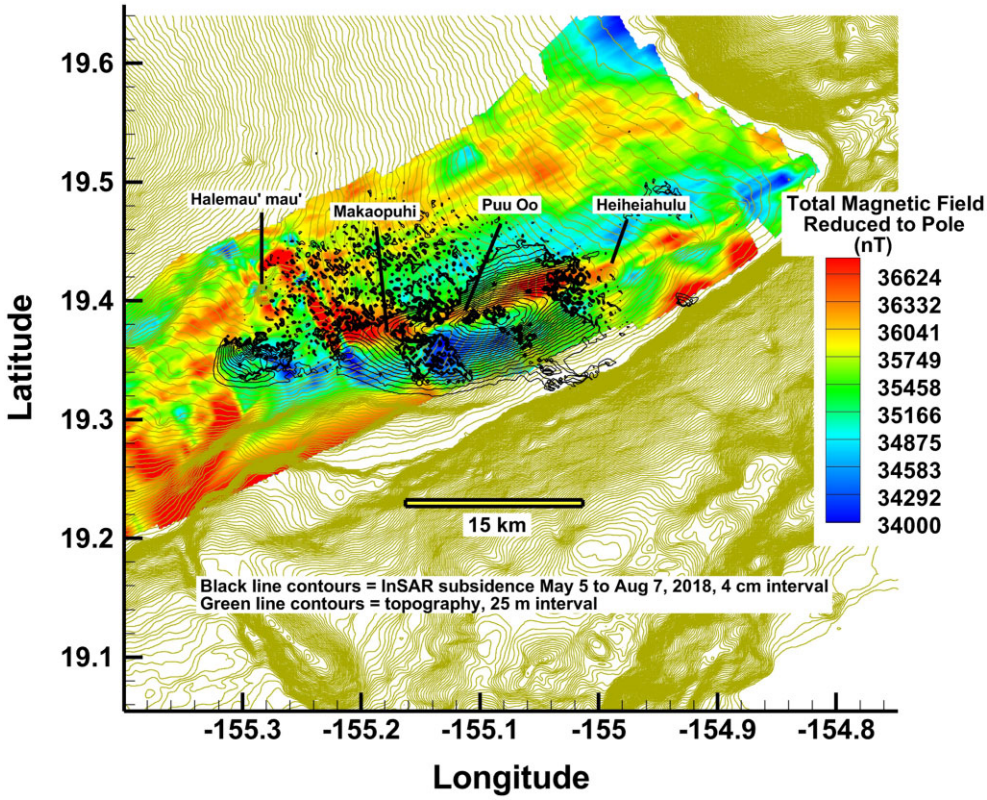

Figure 2. Geological and geophysical data support the continuity of the magma system from the summit into the ERZ on the Big Island of Hawai'i (Hildenbrand *et al.* 1993; Kauahikaua *et al.* 2000; Flinders *et al.* 2013). Shown here is the total magnetic field reduced to the pole (RTP), in which the nearly continuous magnetic high from the summit and out along the ERZ records the magnetic field of dykes emplaced at the summit and along the rift zone that geodesy studies show originate from sources below 3 km depth along the rift zone (Lundgren *et al.* 2013). Gravity data form a similar pattern, and this is a ubiquitous feature of the summits and rift zones of Hawaiian volcanoes that likely reflects the accumulation of olivine beneath regions of magma storage as the volcano grows (Clague & Denlinger 1994; Clague & Sherrod 2014). Also shown is the unwrapped InSAR-derived subsidence (descending sentinel data 2018 May 5 to August 9; d$e = 0.545$, d$n = 0.118$ and d$z = 0.830$) during extraction of 1.5 km^3 DRE of melt from Kīlauea's magma system (Dietterich *et al.* 2021).

Table 1. New coefficients in the modified Nafe–Drake velocity–density equation

$\bar{w}_0\|_{\text{prior}}$	$w_0\|_{\text{posterior}}$	w_1	w_2	w_3	w_4	w_5
278	170	1661.2	−472.1	67.1	−4.3	0.106

where the weights w_i are given in Table 1, and the basis functions $\phi_i(\mathbf{v})$ are given by

$$\phi_i(\mathbf{v}) = v_p^i. \tag{2}$$

Here $\mathbf{v}(x, y, z)$ is the velocity at a point within a 3-D subsurface domain. In describing their laboratory measurements of the relationship between compressional velocity and density, Manghnani & Woollard (2013) shifted the Nafe–Drake curve shown in Fig. 3 upwards by adding a constant bias of 220 kg m^{-3}. This bias is included in eq. (1) as

$$\rho(\mathbf{v}) = \sum_0^5 w_i \phi_i(\mathbf{v}) = \mathbf{w}^{\mathbf{T}} \boldsymbol{\phi}(\mathbf{v}) \tag{3}$$

in which the basis function for the bias at $i = 0$ is simply unity. We can fit this equation to any set of data using a Bayesian linear regression. Here, we determine a prior model for the bias alone, from the lab data of Manghnani & Woollard (2013), assuming the remaining weights are fixed. This determines a prior for the bias and its uncertainty based on lab data. Then we determine the maximum likelihood for the bias using field data. We use a combination of gravity disturbance data and seismic tomography as described in Denlinger & Flinders (2022), with a Gaussian noise model for both sets. We then obtain the maximum posterior for the bias from the product of the lab-determined prior and field-determined maximum-likelihood distributions. As they are both Gaussians, their product is also Gaussian.

The distribution of bias values for the laboratory data of Manghnani & Woollard (2013) about their mean is a smoothly varying, symmetric distribution, and the mean value is approximately the mode value, supporting a Gaussian model for the noise distribution. The likelihood of observing a density ρ from M laboratory measurements of the density and velocity is the conditional probability

$$p(\rho|w_0, \mathbf{v}, \sigma_0^2) = \prod_1^M \mathcal{N}(\rho_m|w_0, \mathbf{w}^{\mathbf{T}}\boldsymbol{\phi}(\mathbf{v}), \sigma_0^2) \tag{4}$$

in which $\mathcal{N}$ indicates a Gaussian distribution, σ_0^2 is the variance determined from the scatter in the data and w_0 is a particular value for the bias, and the other weight parameters (the Nafe–Drake distribution) are fixed. The error function $E(w)$ obtained from the log of this expression is

$$E(w) = \frac{1}{2}\sum_{m=1}^{M}\left[\rho_m - w_0 - \sum_{j=1}^{5} w_j \phi_j\right]^2. \tag{5}$$

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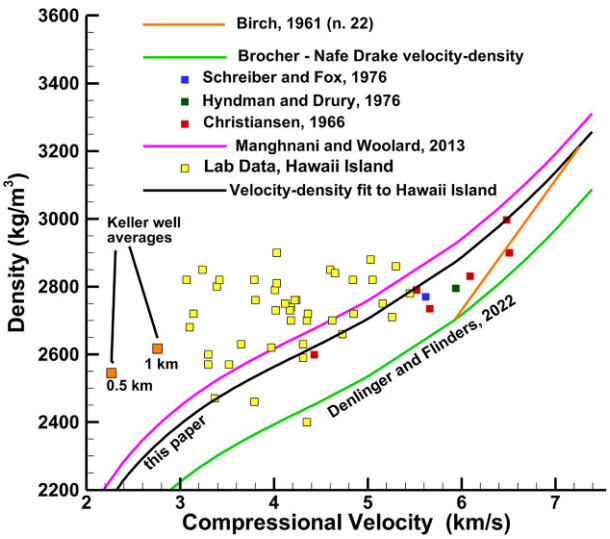

Figure 3. Laboratory measurements for mafic rocks (Birch 1961; Christensen 1966; Hyndman & Drury 1976; Manghnani & Woollard 2013) are used to convert compressional seismic velocity to rock density in order to interpret the density structure of the Earth's crust and mantle. Here-, some of these data are combined with similar measurements of mafic Hawaiian rocks. The purple curve was determined by Manghnani & Woollard (2013) to provide a least-squares fit to lab data for Hawai'i, and is the velocity–density relationship of Nafe–Drake plus a constant, velocity-independent bias of 220 kg m^{-3}. In this paper, we determine that that bias for gravity and seismic tomography data island-wide is 170 kg m^{-3}.

Minimizing $E(w)$with respect to the bias w_0 (Bishop 2009) gives a value for the mean bias $\bar{w}_0$

$$\bar{w}_0 = \bar{\rho} - \sum_{1}^{5} w_i \bar{\phi}_i \tag{6}$$

in which

$$\bar{\rho} = \frac{1}{N}\sum_{1}^{N} \rho_n \tag{7}$$

and

$$\bar{\phi}_i = \frac{1}{N}\sum_{1}^{N} v_p^i. \tag{8}$$

These values are used as a prior for the bias value w_0 in a Bayesian analysis of the field data, and its normal distribution is written as

$$p(w_0) = \mathcal{N}(w_0|\bar{w}_0, \beta_0) \tag{9}$$

in which for an uninformative prior for the bias values, the inverse of the variance β_0 is estimated by the mean-squared difference between the Manghnani line and the lab data in Fig. 3.

In determining the constraints from the observed gravitational field we use the gravity disturbance values calculated as described and explained in Denlinger & Flinders (2022) and Damiani (2013). The gravity disturbances used here are the differences between the observed gravity field at a point minus the normal gravity of the WGS84 geoid that passes through that point. Additionally, these disturbances have been corrected for the flexure of the lithosphere, as described in Denlinger & Flinders (2022). The result is a gravity disturbance field due solely to the 3-D distribution of mass in the island edifice, whether the mass exists above or below the measurement point. This is not a Bouguer anomaly, which subtracts an arbitrary value from a measured field to obtain a residual and in which it is often assumed that all density bodies lie below the measurement point.

The likelihood used here is the probability that we will observe a gravity disturbance value g_n at a surface location $\mathbf{x}(x, y, z)$ resulting from a velocity-determined density distribution of the edifice. The likelihood function for the gravity field data $\mathbf{d}(x, y, z)$, obtained from surface gravity measurements with an assumed Gaussian noise distribution, is the product of conditional Gaussians, one for each field point

$$p(\mathbf{d}|\mathbf{v}, \mathbf{w}, \beta) = \prod_{1}^{N} \mathcal{N}(g_n | f_n\left(\mathbf{w}^{\mathbf{T}}\boldsymbol{\phi}(\mathbf{v})\right), \beta^{-1}). \tag{10}$$

Here, N is the number of surface field points at which we determine the gravity disturbance for the edifice at that point, using the method presented in Denlinger & Flinders (2022). The term $f_n\{\mathbf{w}^{\mathbf{T}}\boldsymbol{\phi}(\mathbf{v})\}$ is the function that determines the surface gravity disturbance at location n from a hexahedral prism whose centroid has a velocity v_p at some subsurface location in the edifice. The model gravity is subject to the subsurface estimate of velocity-determined density, which has a Gaussian uncertainty with precision (inverse variance) β. The error function $E_D(w)$ is obtained from taking the log of eq. (11) and has a structure similar to eq. (5), and

$$E_D(w) = \frac{1}{2}\sum_{n=1}^{N}\left[g_n - f_n\left\{\mathbf{w}^{\mathbf{T}}\boldsymbol{\phi}(\mathbf{v})\right\}\right]^2. \tag{11}$$

The density of each subsurface hexahedral prism contributing to the surface gravity disturbance field is determined from its average compressional velocity, using the 3-D seismic tomography structure of Lin *et al.* (2014) and eq. (5) for any constant bias value w_0. The mesh used for this regression is described in Denlinger & Flinders (2022). Minimizing this error function for each hexahedral prism contributing to the gravity disturbance at the surface point (for the entire island edifice) gives the maximum-likelihood value for the bias from all the field data. As will be shown below, since the number of field measurements N greatly outnumbers our single parameter, then the inverse variance is approximately given by

$$\beta = \frac{N}{2E_D(w)}. \tag{12}$$

The prior is given by $p(w_0)$ (eq. 9) and combined with the likelihood determines the unnormalized Bayesian posterior for the bias w_0 as

$$p(w_0|\mathbf{x}, \mathbf{w}, \sigma) \propto p(\mathbf{d}|\mathbf{x}, \mathbf{w}, \sigma)p(w_0) \tag{13}$$

By completing the square in the exponentials in this product (Bishop 2006, p. 93), and normalizing this Gaussian, the product on the right-hand side can be written as (Bishop 2009, p. 153),

$$p(w_0|\mathbf{d}) = \mathcal{N}(w_0|\mathbf{m}_N, \mathbf{S}_N), \tag{14}$$

where

$$\begin{aligned}\mathbf{m}_N &= \mathbf{S}_N(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\phi}^T\mathbf{d})\\ \mathbf{S}_N^{-1} &= \mathbf{S}_0^{-1} + \beta\boldsymbol{\phi}^T\boldsymbol{\phi}.\end{aligned} \tag{15}$$

Here $\mathbf{m}$ refers to the mean (and mode) of the distribution and $\mathbf{S}$refers to the covariance. The subscript 0 refers to the prior lab data, whereas Nrefers to the posterior for bias w_0 using Nfield points. These statistics are a quantitative measure in the uncertainty in our density structure at depth, as the variations in bias w_0 determine the variations of density for each value of velocity at a point. Eqs (14) and (15) are a general result for a range of possible weight bias values

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Table 2. Sufficient statistics for prior and posterior Gaussians for bias.

Mean prior $\bar{w}_0$	Variance prior σ_0^2	Mean bias $\bar{w}_0$	Variance σ^2
278.	158^2	170	22^2

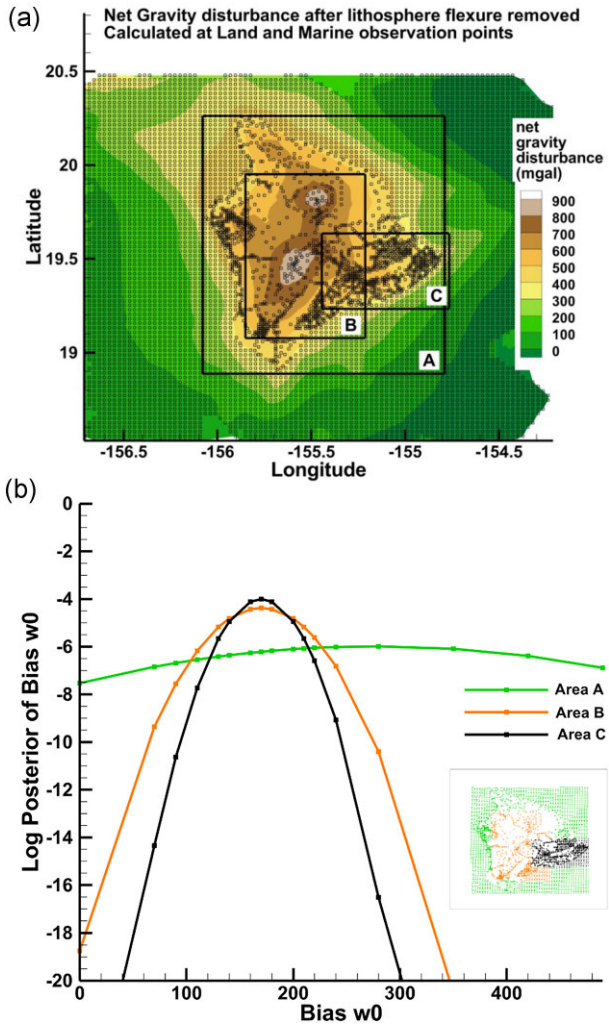

Figure 4. (a) Three different areas used to test the efficacy of the bias generalizing the velocity–density relationship for Hawaiian rocks. The area A is what was used in Denlinger & Flinders (2022), whereas the areas B and C are confined mostly to subaerial locations best constrained by the seismic networks used to do tomography (Lin *et al.* 2014). The statistics for A, B and C (see the text) are listed in Tables 1 and 2. (b) The posterior distribution for the bias w_0 determined for each area in (a). As discussed in the text, the consistency of the posterior for land areas B and C, despite large differences in the area of these two domains (see inset), is a robust result. For Kilauea volcano (curve C), the standard deviation in the maximum posterior distribution for the bias is 22 kg m^{-3} (Table 2).

and Gaussian errors in data and measurements. For this problem here, we have held fixed all of the weights given by Brocher (2005), and only allow the bias to vary. In this case, for the posterior distribution over the single variable for bias w_0, eqs (14) and (15) are reduced to:

$$p\left(w_0|m_N, \sigma_N^2\right) = \mathcal{N}\left(w_0|m_N, \sigma_N^2\right) \tag{16}$$

in which

$$m_N = \frac{\sigma^2}{N\sigma_0^2+\sigma^2} m_0 + \frac{N\sigma_0^2}{N\sigma_0^2+\sigma^2} m$$
$$\frac{1}{\sigma_N^2} = \frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}. \tag{17}$$

Here $\sigma^2 = \beta^{-1}$ and $\sigma_0^2 = \beta_0^{-1}$ are the variances in the fit to the gravity disturbances and the fit to the prior lab data, respectively. The uncertainty in density is directly given by the uncertainty in the maximum in the posterior for bias w_0. The probable variation in density around the maximum posterior used in the remainder of this paper is given by the variance for the bias listed in Tables 1 and 2.

Consistency in these results, and therefore this posterior for the bias, depends upon where sampling is done. Given that the seismic constraints on the subsurface were confined to seismometers on land (Lin *et al.* 2014), the constraints for velocity in the subsurface within the volcanic edifice, for example, above decollement seismicity, are strongest beneath land rather than the seafloor. In Fig. 4(a), the gravity disturbance field determined by Denlinger & Flinders (2022) is shown, along with three areas sampled to determine the posterior probability of the constant bias w_0. The corresponding normalized posterior distributions, obtained from the product of the likelihood for that area and the prior Gaussian determined from the lab data, are shown in Fig. 4(b). The largest area A coincides with the lateral margins of the grid used by Lin *et al.* (2014) for seismic tomography. The next largest area B includes the summits of both Mauna Loa and Mauna Kea volcanoes. The smallest area C is confined largely to Kīlauea volcano, and is less than half of the area of B. Nevertheless, both B and C have the same maximum posterior for the bias parameter w_0. These two smaller areas are primarily confined to land, and there the tomographic velocity structure is much better constrained than it is offshore where no seismometers were deployed. The values for the mean and variance of the posterior for the bias parameter for area C is listed in Table 2. The resulting equation, formed from the weights in Table 1 and the posterior bias in Table 2, form the basis for a new density structure for Kīlauea volcano. The gravity disturbance residuals obtained by applying the maximum posterior bias of 170 kg m^{-3} to convert velocity to density are shown in Fig. 5. As shown in Fig. 4(b), the error obtained from this bias is significantly better than the value of 220 kg m^{-3} that Manghnani & Woollard (2013) used, and a marked improvement over the zero bias initially used by Denlinger & Flinders (2022) that is shown by the green curve in Fig. 4(b).

DISCUSSION

The new velocity-derived density structure significantly increases densities within the previously determined density structure of Denlinger & Flinders (2022) and provides a new structural framework for Kīlauea volcano. Shown in Fig. 6 is a view of dense 3-D isosurfaces within Kīlauea volcano and in the underlying oceanic lithosphere that have the density of dunite (3220 kg m^{-3}), and these are combined with the relocated seismicity (Lin *et al.* 2014) associated with either fault movement or magma transport into the magma system. The seismic hypocentres, shown as black dots, form distinct spatial groups identified by letters in this figure. Letter A is left

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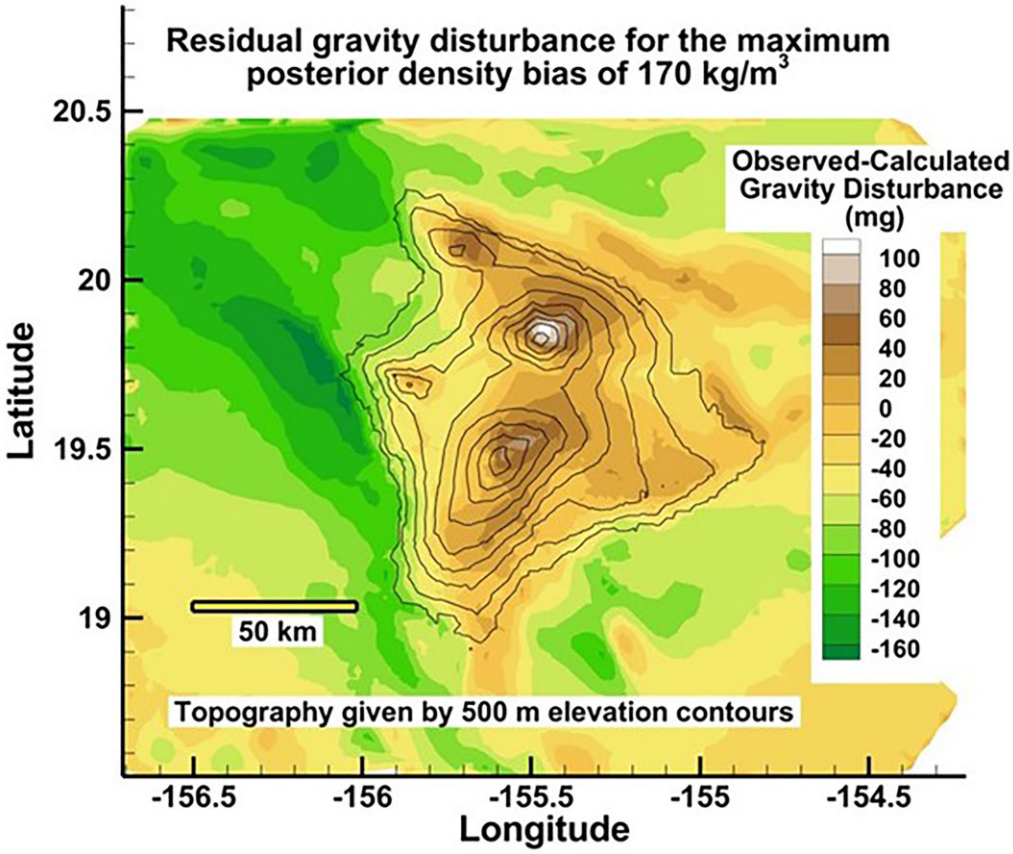

Figure 5. The distribution of the residual gravity disturbance in which the calculated values used the seismic tomography of Lin *et al.* (2014) to construct the 3-D domain of Denlinger & Flinders (2022). The linear regression used the velocity–density relationship in eq. (3) with the values in Tables 1 and 2, and the maximum posterior bias w_0 shown in Fig. 4(b) and given in Table 2. Observed gravity disturbances at each surface point result only from the island edifice structure, as described in Denlinger & Flinders (2022), and are modelled with a 3-D model of the island edifice to produce the residuals shown here.

lateral strike-slip seismicity along the upper ERZ of Kīlauea (Gillard *et al.* 1996), and flanks the east side of the south caldera magma source interpreted from geodetic studies (Poland *et al.* 2012). Letter B is right lateral strike-slip and wrench fault seismicity occupying the west half of the Koae desert (Lin & Okubo 2016). Seismicity labelled C is produced largely by dyke injection within the summit caldera. Letter D is right lateral plus extensional faulting induced by the S20°E motion of the south flank of Kīlauea as it slides into the sea (Denlinger & Okubo 1995; Owen *et al.* 2000; Wyss *et al.* 2012). Letter E is the path of melt migration through the oceanic lithosphere and into Kīlauea volcano, as illuminated by the earthquakes that melt migration induces (Klein *et al.* 1987). The track of this seismicity suggests that melt migrates upward into and through the large, dense (3220 kg m^{-3} isosurface), body underlying the nearly aseismic region between the seismicity C and that of A and B. This aseismic summit region contains the sources of pressure change that characterize summit eruptive activity and summit surface deformation on Kīlauea volcano (Anderson *et al.* 2015; Wang *et al.* 2021).

Within the aseismic summit region in Fig. 6, there are density structures above −5 km elevation and below −2.5 km elevation that cradle the magma system. To illustrate this, we show the dense bodies in Fig. 6 in map view in Fig. 7(a), along with topography and epicentres of seismicity, and give the orientation of the vertical SW-NE cross-section in Fig. 7(b). This vertical slice is constructed to bisect the Halema'uma'u (HMM) collapse, the deeper seismicity associated with magma transport into the base of the volcano, and the shallow HMM magma source of Wang *et al.* (2021). This section is viewed looking north in Fig. 7(b), and shows that density isosurfaces between densities of 2600 and 3000 kg m^{-3} near HMM form a cupola at shallow depths that is contained within seismicity C, the aseismic region, and then inverts for form a trough as depth increases. The cupola protruding into the shallow layers envelopes the ellipsoidal HMM pressure sources of both Wang *et al.* (2021) and Anderson *et al.* (2015), and at 2600 kg m^{-3} is significantly denser than the topmost km of basalt (Keller *et al.* 1979). For reference to Fig. 6, we show the hypocentre distributions of earthquakes projected onto this plane, labelled with the same labels as in Fig. 6. In the foreground of Fig. 7(b) cross-section, strike-slip seismicity is plotted that flanks the deeper trough, and is consistent with melt and mush flow south out of the summit region.

In three dimensions, the density isosurfaces produced by the contours shown in the cross-section of Fig. 7(b) form a definitive structure at the summit of Kīlauea volcano. In Fig. 8(a) suite of these surfaces is shown relative to the continued subsidence of the summit immediately following the cessation of the massive 2018 eruption. For three months, melt and mush continued to flow out of the summit area and into the ERZ, which underwent concomitant uplift along the rift just east of Pu'u'ō'ō (Wang *et al.* 2021). Deflation of the south summit area, considered to be the main storage

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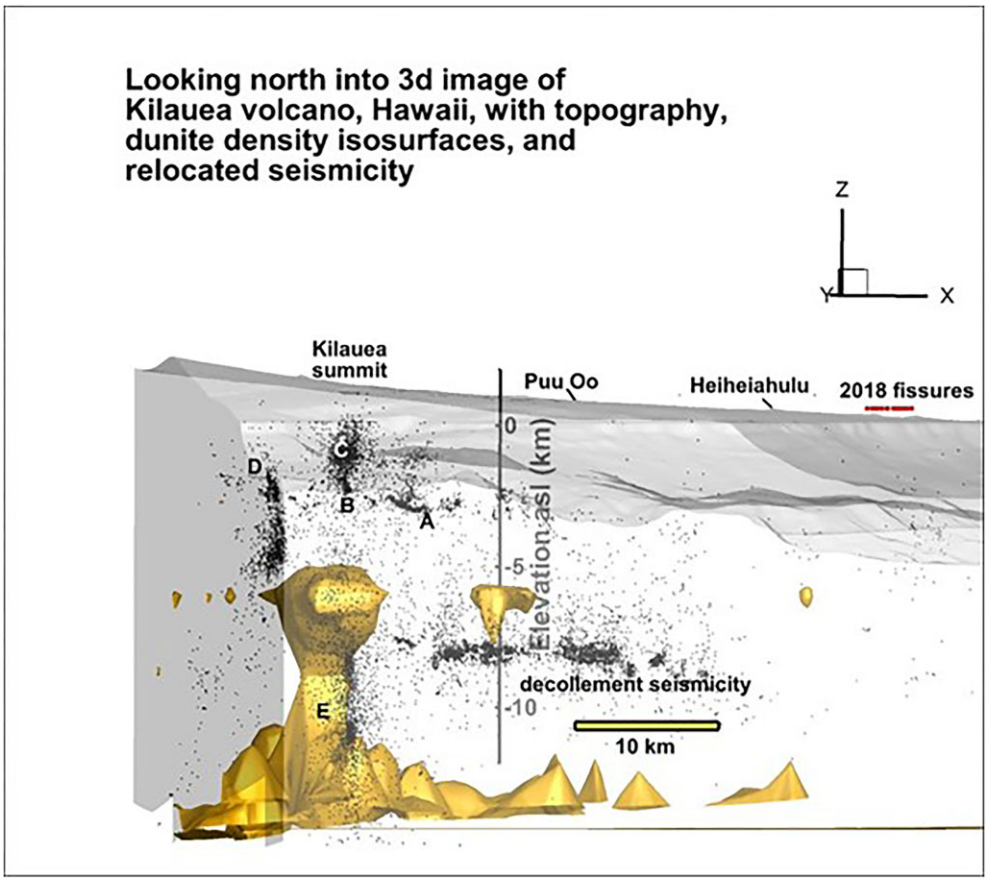

Figure 6. The density structural model derived from seismic tomography (Lin *et al.* 2014) and the gravity disturbance field (Denlinger & Flinders 2022) produces a structural framework for Kīlauea volcano in which dense bodies underlie the locations of magma storage inferred from geodetic studies of eruption-related surface deformation. This is true both at the summit, and along the ERZ beneath Makaopuhi (not shown), Pu'u'ō'ō, and Heiheiahulu. Here, decollement seismicity defines the base of the edifice sliding into the sea, and the other letters indicate persistent seismic patterns. Letter A is left lateral seismicity (Gillard *et al.* 1996) and letter B is right lateral and wrench faulting seismicity (Lin & Okubo 2016) that flanks the location of a geodetically determined south caldera magma source (Wang *et al.* 2021), on the east and west sides, respectively. The letter C is shallow summit seismicity between 0.5 and −2 km elevation. The letter D is right lateral wrench faulting on Mauna Loa, produced as Kīlauea slides SE into the sea (Wyss *et al.* 2012). Letter E is generated by magma movement from the mantle, through the oceanic lithosphere and into the volcano, passing through the dense body underlying the summit region. The 2018 fissures are where 1.5 km^3 of melt drained in 60 d from the volcano in 2018 (Dietterich *et al.* 2021), causing a caldera collapse at the summit.

area for melt and mush at the summit (Anderson & Poland 2016), has a surface expression with a curvature and with inflections that coincide with a range of density isosurfaces between 2800 and 2900 $kg\,m^{-3}$ (Fig. 8). In particular, the sharp inflection in subsidence contours and the shape of the depression south and west of the summit are mimicked by the tomography-determined density structure at depth. The closest correspondence is with a density of 2880 $kg\,m^{-3}$, as shown in Fig. 8.

The density isosurface that most closely corresponds to the post-eruption 2018 August–November summit subsidence contours (2880 $kg\,m^{-3}$) also has structure that aligns with summit seismicity, with known vents along the ERZ, and with the subsidence observed along the ERZ during the large 2018 eruption described by Neal *et al.* (2019). The correspondence of this density isosurface with summit and Koae seismicity is shown in Fig. 9, along with line contours of topography. The trough underlying shallower magma sources in the summit opens to the south, as shown by the flooded contour surface here, and this opening is flanked by two distributions of strike-slip earthquakes, one left lateral along the upper ERZ, and one right lateral extending south from the crater area as shown [the latter distribution is wrench faulting (Mandl 1988) as it also includes extensional mechanisms]. This pattern is also reflected in the deformation of the Koae fault system over hundreds of years (Duffield 1975). The strike-slip seismicity is likely induced by flow of melt and crystals south out of the summit towards the ERZ. Along the ERZ, elevated 'hills' in this density isosurface represent higher densities, and prominent hills and ridges underlie known vents at Makaopuhi, Pu'u'ō'ō and Heiheiahulu along the ERZ. In Fig. 10, this 2880 $kg\,m^{-3}$ isosurface is shown with the underlying dense 3220 $kg\,m^{-3}$ isosurfaces forming the structures shown in Fig. 6. These are associated with magma storage in the summit, and we infer that the same is true along the ERZ. Along the portion of the 2880 $kg\,m^{-3}$ surface in the ERZ, there is a sinuous channel formed by the elevations in this surface which also corresponds to the subsidence observed during the 2018 eruption, as shown in Fig. 11.

All of the isosurfaces between 2800 and 2900 $kg\,m^{-3}$ form an elevated, roughly triangular-shaped platform formed by the plunge of these isosurfaces to depths of 6 km or more along the platform margins (Fig. 9). This plunge in the elevation of these isodensity

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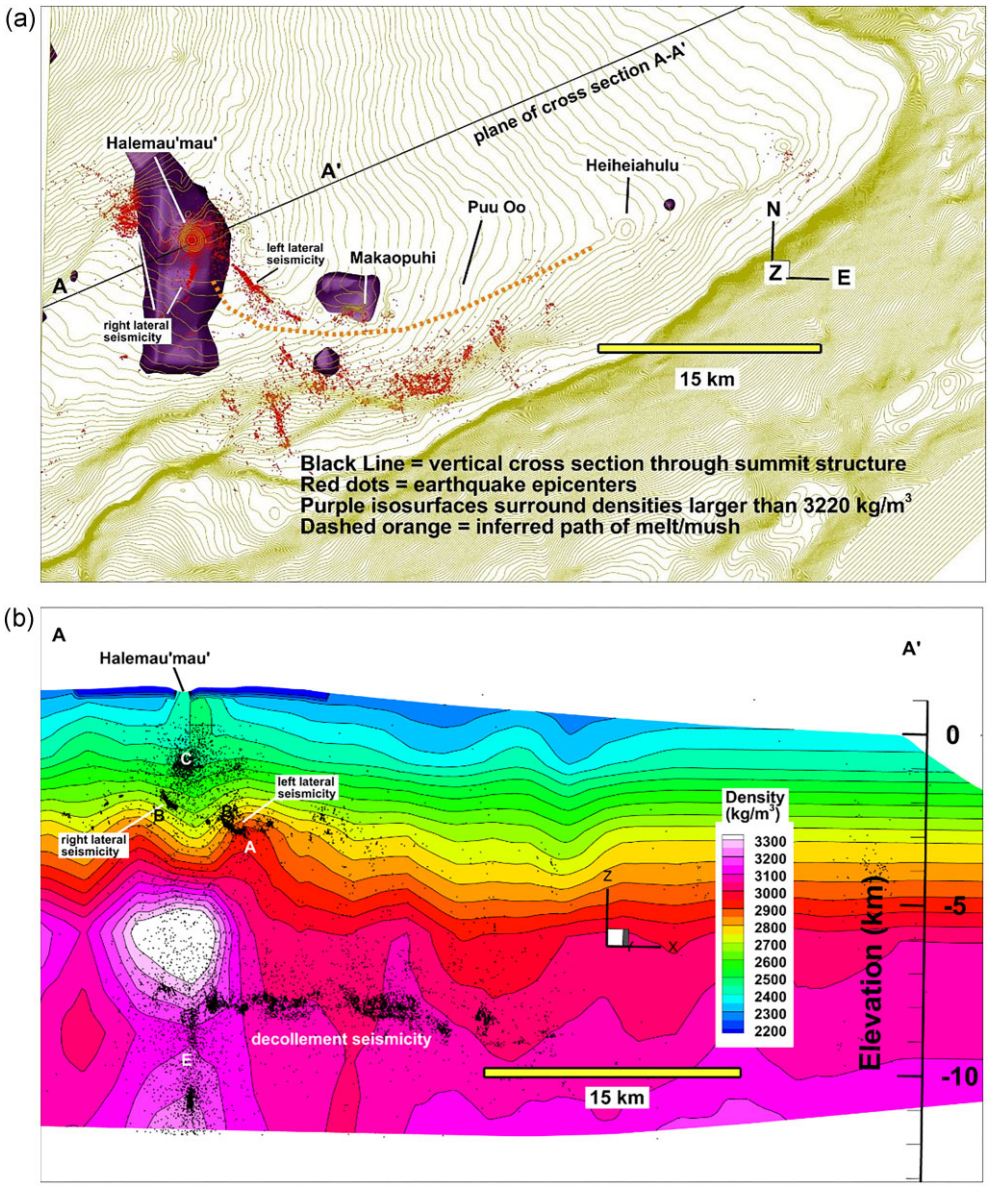

Figure 7. (a) Map view of dense bodies in Fig. 6, with seismicity and orientation of vertical cross-section in (b). The cross-section intersects isosurfaces that define the summit density structure above the dense 3220 kg m^{-3} bodies shown here, and these form the trough that is flanked by strike-slip seismicity in (b). (b) Vertical cross-section through seismic tomography derived density structure that intersects the summit 3-D body and seismicity of C shown in Fig. 6. Seismicity outside this plane are projected horizontally onto it. This section shows a trough beneath Halema'uma'u at Kīlauea's summit that often produces sources of inflation and deflation. These sources are inferred to be a mixture of melt and crystals, and their flow south out of the summit region are consistent with distribution of out-of-plane strike-slip seismicity shown here. The 3-D shape of this trough for various densities is compared to InSAR derived summit subsidence in Fig. 8.

surfaces indicates a substantial reduction in density surrounding this platform. To the south, these soft and less dense areas closely correspond to the arcuate boundaries of the Hilina Pali fault system (Fig. 9). These reduced densities around the margins of Kīlauea volcano may represent a reduction in density produced by hydrothermal alteration, as these anomalies flank the active, hotter, subsurface areas of Kīlauea volcano. If so, then the Hilina Pali faults are rooted in weak, hydrothermally altered material formed by hydrothermal circulation flanking the existing magma system, and their formation is exacerbated by S20E extension of the south flank without being directly connected to decollement slip (Denlinger & Morgan 2014).

Combined with other geophysical and geodetic data these observations form a structural framework for Kīlauea volcano. The accumulations of dense bodies with the density of dunite are likely to be accumulations of olivine that are forming dunite in areas of long-term melt and mush accumulation within Kīlauea's magma system (Lerner *et al.* 2024). These dense bodies are interpreted here to be overlain by a dense mixture of melt and crystals that feed dykes, both at the summit and along the ERZ as shown in Fig. 12(Upper Panel). The density structure of the mush (Fig. 7b) is consistent with concentration of olivine crystals increasing with depth, and this gradation in density along the ERZ is supported by

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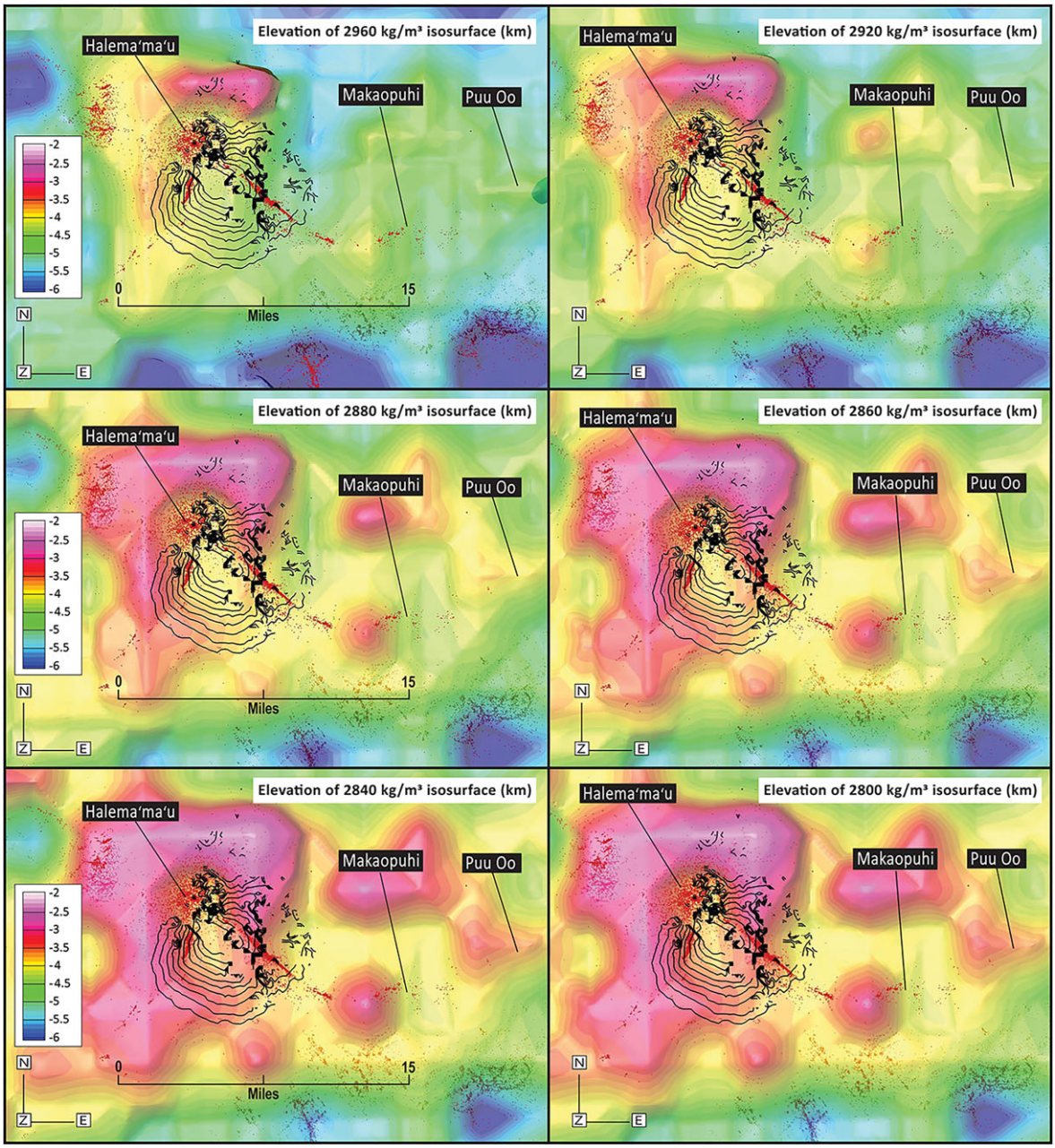

Figure 8. Map view of density isosurfaces within the aseismic region overlying the dense bodies in Fig. 6. Black line contours show InSAR subsidence as summit continued to drain from 2018 August 9 to November 7. In addition, red dots are earthquake hypocentres above the surface, and purple dots are earthquake hypocentres below the surface. Flooded contours show a suite of these surfaces relative to both seismicity and to the InSAR-derived subsidence in the summit (unwrapped descending sentinel data 2018 August 9 to November 7 with look angle components $de = 0.545$, $dn = 0.118$ and $dz = 0.830$) as the summit drained into the ERZ from 2018 August to November. Close correspondence between bowl-shaped density isosurfaces and these contours in InSAR-derived vertical displacement in the south summit area occurs for densities between 2880 and 2840 kg m^{-3}. Melt density is about 2600 kg m^{-3}.

the 1840 May–June eruption along the ERZ, where crystal poor lava was erupted near Makaopuhi and picrite with cm-size olivine crystals was erupted just uprift of Cape Kumukahi (Macdonald 1944). It is also the structure of spreading ridges that form in backarc basins (Turner *et al.* 1999; Veloso *et al.* 2014; Artemieva 2023). It is also the structure proposed by Walker (1986) in his analysis of the Koolau dyke complex on Oahu. It is the correspondence of Hawaiian ridge structure and growth with oceanic and spreading ridge structure that is the basis for Fig. 12.

At the summit, our data density is sufficient to indicate that the density and thus crystal concentration increases nonlinearly with depth, eventually forming the dunite bodies that underly this region of magma storage within the volcano. Here, and along the ERZ, a combination of variations in melt concentration and the separation of the flow of melt from a crystalline mush to feed dykes and lava flows can produce complex mechanical behaviour (Suckale *et al.* 2016; Keller & Suckale 2019; Bessat *et al.* 2022). This applies to melt transport during periods between eruptions as well as during

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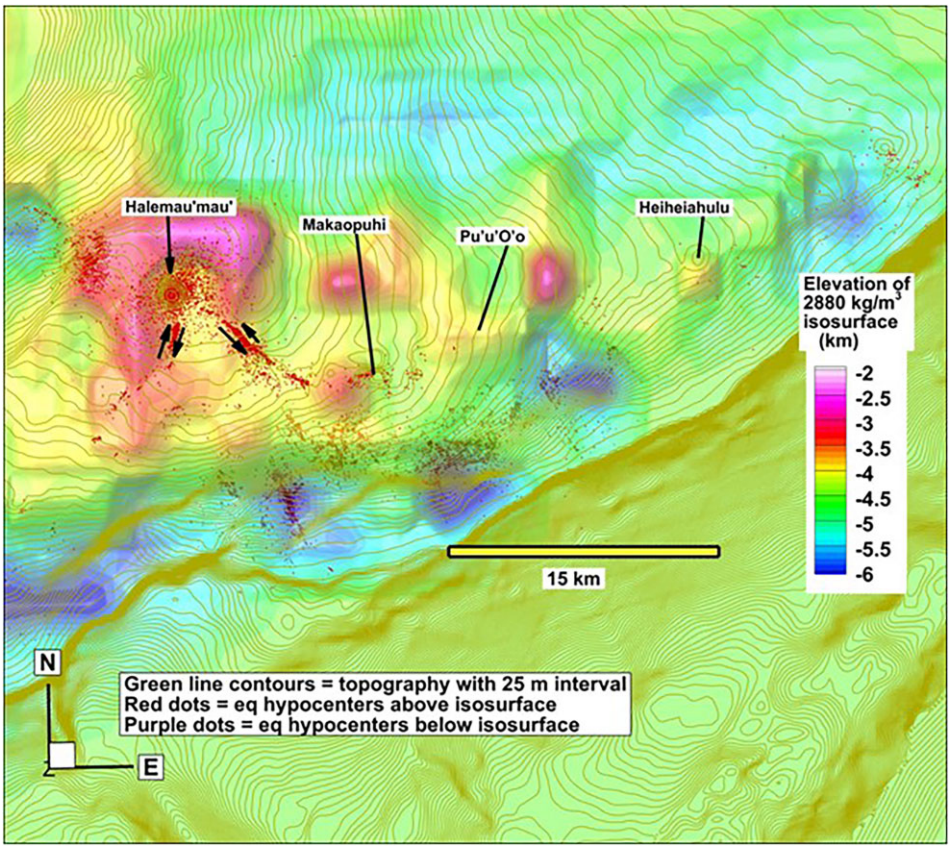

Figure 9. Map view of the topography and seismicity associated with the 2880 kg m^{-3} density isosurface that most closely aligns with the shape of south summit subsidence as melt and mush flowed out of the summit from 2018 August 9 to November 7 (Fig. 8). This flow of melt is consistent with observed seismic focal mechanisms of strike-slip seismicity flanking this trough as shown here. Elevated 'hills' in this density isosurface are associated with past eruptive vents along the ERZ, such as near Makaopuhi, Pu'u'ō'ō and Heiheiahulu. The roughly triangular region formed by these elevated areas (shown here as yellow and warmer colours), is bounded by areas shown in blue where this isosurface plunges to depths of 6 km or more. The low density of the edifice may reflect hydrothermal circulation surrounding this hot region. These regions of low density align with the Hilina Pali fault zone, defined by the prominent, arcuate scarps in this figure, and if generated by hydrothermal alteration indicate that these fault scarps are rooted in weak hydrothermally altered material.

eruptive activity. Though these complex behaviours are beyond the scope of this study, they provide numerous opportunities for future research on this system.

CONCLUSIONS

The relationship between compressional velocity and density as given by the Nafe–Drake relationship (Brocher 2005) requires a constant bias to explain joint measurements of seismic velocity and density for Hawaiian rocks (Manghnani & Woollard 2013). Allowing for a constant bias, the 3-D seismic tomography structure can be converted to a 3-D density structure for the island of Hawai'i, and produces a surface gravity disturbance field that more accurately reproduces the observed gravity disturbance field on the island than was done previously. Using the laboratory measurements as a prior, a linear Bayesian regression of thousands of gravity disturbance field points on land and at sea determines a maximum posterior for this bias of 170 kg m^{-3}. This constrains the 3-D density structure of the island, and particularly of Kīlauea volcano.

The new density structure defines dense bodies at shallow levels at the summit and along the ERZ of Kīlauea volcano that have the density of dunite (3220 kg m^{-3}). These dense bodies underlie subsurface regions that geodetic and geophysical studies routinely identify as volumes of pressure change that produce observed surface deformation, and therefore are correlated with magma storage. However, we show here that the density of these regions is greater than the density of a crystal-free tholeiitic basaltic melt, and here the volumes where pressure change occurs are interpreted as a melt-rich layer overlying a crystal rich mush. The bulk density increases nonlinearly over a depth range of 3–4 km, culminating in bodies with the density of dunite (>3220 kg m^{-3}). At the summit, density isosurfaces between 2750 and 3000 kg m^{-3} form a horse-collar-shaped trough that opens to the south and cradles the magma volumes associated with surface deformation during volcanic unrest. In map view, contours of this trough surround both the LIDAR-determined 2018 summit collapse (Mosbrucker *et al.* 2020) and also aligned with the continued subsidence of the south caldera region after the end of the 2018 eruption. On its exit from the crater, the south trending 2880 kg m^{-3} density isosurface forms a trough that is flanked by strike-slip seismicity. Focal mechanisms of this seismicity are consistent with the southern flow of a mush mixture south and east out of the crater and into the ERZ. Along the ERZ, the 2880 kg m^{-3} isosurface has hills that underlie previously active vents at Makaopuhi, Pu'u'ō'ō and Heiheiahulu. The axis of a sinuous trough snaking between these hills aligns with InSAR-determined ERZ subsidence,

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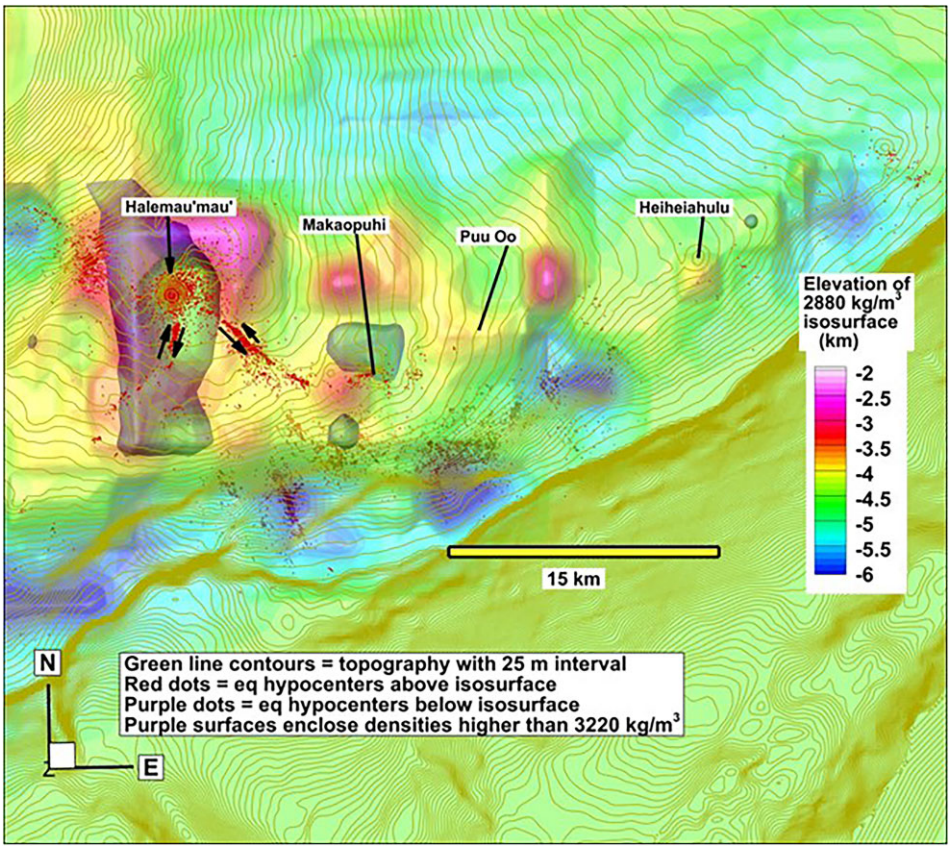

Figure 10. Map view of the relationship of the 2880 kg m^{-3} isosurface shown in Figs 8 and 9 with the dense bodies shown in Figs 6 and 7(a). Green lines are topographic contours with a 25 m interval. Dots are earthquake epicentres, with red above the surface and purple below. Darker isosurfaces enclose high densities below flooded contour surface. The density of these bodies (3220 kg m^{-3}) is consistent with dunite and is likely an olivine cumulate derived from present or past magma storage within the volcano. These bodies, also mapped by magnetotellurics (Hoversten *et al.* 2022), are largest under the summit, and are consistent with picrites found in the summit region (Casadevall & Dzurisin 1987). This structure, through which the magma entering the volcano passes (Fig. 6), may be a common byproduct of the growth of Hawaiian volcanoes (Clague & Denlinger 1994; Clague & Sherrod 2014). It provides a mechanism for the eruption of golf ball-size nodules of dunite coated in basalt that are found on rift zones of other volcanoes on the island of Hawai'i (Chen *et al.* 1992).

and deflection of that subsidence, during the 2018 eruption. In addition, as these isodensity surfaces between 2800 and 2900 kg m^{-3} correlate to the volcanic activity, the elevated, roughly triangular shaped region within Kīlauea volcano that they form is interpreted to be at elevated temperatures. Surrounding this region, density isosurfaces plunge to depths of 6 km or more and reflect areas of low density surrounding the platform. In particular, these low-density volumes are aligned with the Hilina Pali fault system and indicate that this fault system is underlain by low-density rock. If the density here is produced by hydrothermal alteration, then the Hilina Pali fault system is there because that part of the flank and those faults are rooted in hydrothermally altered material. Taken together, the combination of gravity, seismic velocity, earthquake hypocentres, seismic focal mechanisms and surface deformation induced by volcanic activity show that these derived density structures form a useful skeletal framework for the study of the structure of Kīlauea volcano.

DATA AVAILABILITY

The gravity data underlying this article are available in Flinders *et al.* (2013), at doi:10.1002/grl.50633 and in the USGS Science Base at doi.org/10.5066/F7V1230Q. European Space Agency Sentinel 1 InSAR data are available from Alaska Satellite Facility's data repository (https://asf.alaska.edu/data-sets/derived-data-sets/insar/).

REFERENCES

Anderson, K.R. & Poland, M.P., 2016. Bayesian estimation of magma supply, storage, and eruption rates using a multiphysical volcano model: kīlauea Volcano, 2000–2012. *Earth planet. Sci. Lett.,* **447,** 161–171.

Anderson, K.R., Poland, M.P., Johnson, J.H. & Miklius, A., 2015. *Episodic Deflation-Inflation Events at Kīlauea Volcano and Implications for the Shallow Magma System,* Hawaiian Volcanoes, pp. 229–250.

Artemieva, I.M., 2023. Back-arc basins: a global view from geophysical synthesis and analysis. *Earth Sci. Rev.,* **236,** 104242, doi: 10.1016/j.earscirev.2022.104242.

Bessat, A., Pilet, S., Podladchikov, Y.Y. & Schmalholz, S.M., 2022. Melt migration and chemical differentiation by reactive porosity waves. *Geochem. Geophys. Geosyst.,* **23**(2), e2021GC009963, doi: 10.1029/2021GC009963.

Birch, F., 1961. The velocity of compressional waves in rocks to 10 kilobars—Part 2. *J. geophys. Res.,* **66,** 2199–2224.

Bishop, C.M., 2009. *Pattern Recognition and Machine Learning*. Springer.

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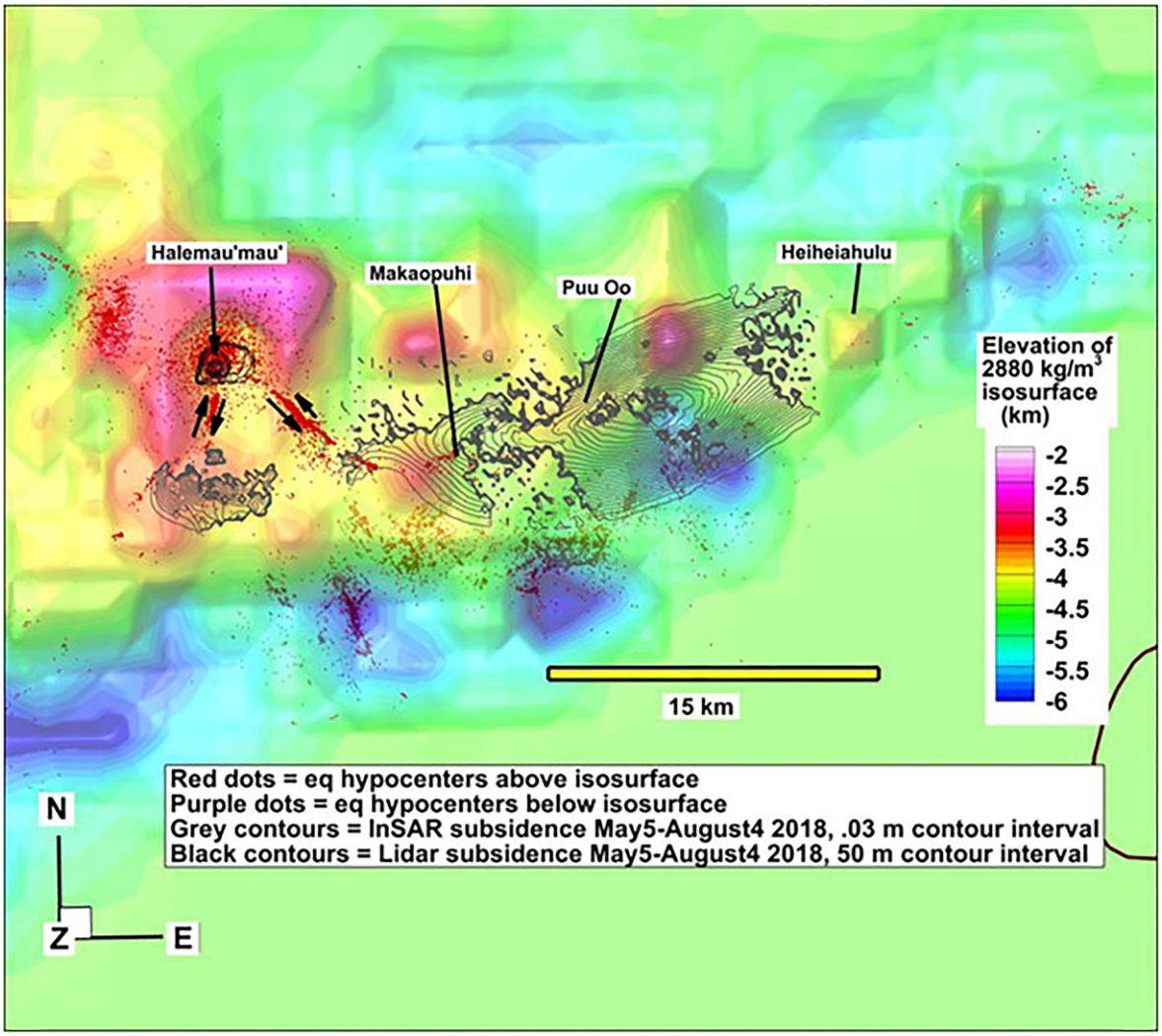

Figure 11. Map view of the relationship of the 2880 kg m^{-3} isosurface shown in Figs 8 and 9 with the large subsidence (grey contours) along the rift zone observed during drainage of 1.5 km^3 of magma (Dietterich *et al.* 2021) from Kīlauea volcano in 2018 (unwrapped InSAR-derived subsidence from descending sentinel data 2018 May 5 to August 9; de = 0.545, dn = 0.118 and dz = 0.830). Black contours show lidar-derived subsidence 2018 May 5 to August 4 in the summit area near Halemau'mau'. The 'horse collar' shape formed by this density isosurface in the summit region, showing higher densities surrounding the active summit caldera, bounds the massive caldera collapse associated with the withdrawal in 2018 that was recorded with lidar, shown in black here overlying summit seismicity in red. The western and southern margins of this horse collar shape bound the limits of InSAR determined subsidence in the summit area. The 'hill' in this surface near Makaopuhi is associated with a deflection in the subsidence mentioned above during 1.5 km^3 of magma withdrawal in 2018.

Brocher, T.M., 2005. Empirical relations between elastic wavespeeds and density in the earth's crust. *Bull. seism. Soc. Am.,* **95,** 2081–2092.

Brooks, B., Foster, J., Bevis, M., Frazer, L., Wolfe, C. & Behn, M., 2006. Periodic slow earthquakes on the flank of Kīlauea volcano, Hawai'i. *Earth planet. Sci. Lett.,* **246,** 207–216.

Casadevall, T.J. & Dzurisin, D., 1987, in Decker, R.W., Wright, T.L. & Stauffer, P.H.(Eds.), *Stratigraphy and Petrology of the Uwekahuna Bluff Section, Kilauea caldera. Volcanism in Hawaii*. U.S. Government Printing Office, Washtington D.C., pp. 351–376.

Chen, C.-H., Presnall, D.C. & Stern, R., 1992. Petrogenesis of ultramafic xenoliths from the 1800 Kaupulehu Flow, Hualalai Volcano, Hawaii. *J. Petrol. ,* **33,** 163–202.

Christensen, N.I., 1966. Elasticity of ultrabasic rocks. *J. geophys. Res.,* **71,** 5921–5931.

Clague, D.A. & Denlinger, R.P., 1994. Role of olivine cumulates in destabilizing the flanks of Hawaiian volcanoes. *Bull. Volcanol.,* **56,** 425–434.

Clague, D.A. & Sherrod, D.R., 2014. Growth and degradation of Hawaiian volcanoes. in: Poland, M.P., Takahashi, T.J. & Landowski, C.M.(Eds.), *Characteristics of Hawaiian Volcanoes*. U.S. Geological Survey. p. 97–148.

Damiani, T., 2013. Gravity disturbances at altitude and at the surface. *Trans. Am. Geophys. Un.,*.

Denlinger, R.P. & Flinders, A., 2022. Density structure of the island of Hawai'i and the implications for gravity-driven motion of the south flank of Kīlauea Volcano. *Geophys. J. Int.,* **228,** 1793–1807.

Denlinger, R.P. & Morgan, J.K., Instability of Hawaiian volcanoes. Chapter 14, in *Characteristics of Hawaiian Volcanoe*. editors: Poland, M.P., Takahashi, T.J. & Claire, M. Landowski U.S Geological Survey Professional Paper 1801, 2014

Denlinger, R.P. & Okubo, P., 1995. Structure of the mobile south flank of Kilauea Volcano, Hawaii. *J. geophys. Res.: Solid Earth,* **100,** 24499–24507.

Dietterich, H.R., Diefenbach, A.K., Soule, S.A., Zoeller, M.H., Patrick, M.P., Major, J.J. & Lundgren, P.R., 2021. Lava effusion rate evolution and erupted volume during the 2018 Kīlauea lower East Rift Zone eruption. *Bull. Volcanol.,* **83,** doi: 10.1007/s00445-021-01443-6.

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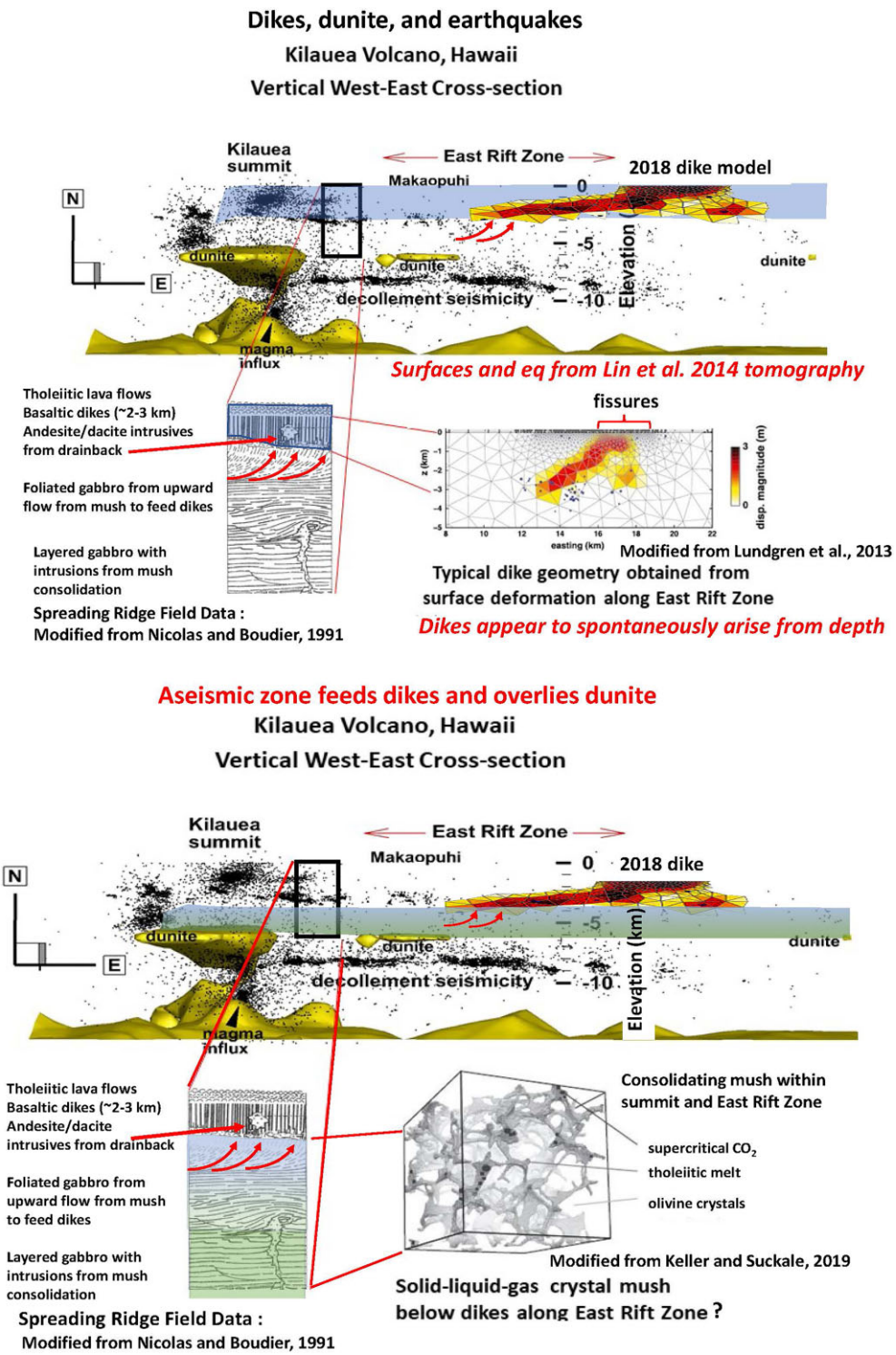

Figure 12. (Upper Panel) Combining geophysical, geological and seismic studies of Kīlauea volcano with the information gleaned here for density, a consistent structural model for the magma system emerges. The structure is similar to the core of spreading ridges that form in backarc basins (Nicolas & Boudier 1991), possibly as a result of the persistent flank motion that produces flank spreading across the ERZ as it migrates seaward, emulating a spreading ridge. Geodetic studies (Montgomery-Brown *et al.* 2010, 2011, 2015; Lundgren *et al.* 2013) show that dykes appear to spontaneously arise from a depth of 3–3.5 km along the ERZ, as shown, and this may simply be a consequence of the presence of extensive, underlying mush bodies that feed them. (Lower Panel) The precipitation and consolidation of olivine crystals form distinct volumes of mush, and produces the dense bodies (labelled 'dunite') that we infer from the combination of seismic tomography, gravity and lab measurements. Variations in melt concentration within this mush are indicated by nonlinear increases with depth, both beneath the summit and along the ERZ, as well as by the near simultaneous eruption of crystal poor magma near Makaopuhi and picrite at the east end of the ERZ (Macdonald 1944). Such variations in melt/mush concentration can produce complex mechanical behaviour (Suckale *et al.* 2016; Keller & Suckale 2019; Bessat *et al.* 2022), particularly in the coupling of the summit to the ERZ during eruptive activity. The potential for a rich set of complex mechanical behaviours resulting from structures like this present abundant opportunities for future research on this system.

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Duffield, W., 1975. Structure and origin of the Koae Fault System, Kilauea Volcano, Hawaii. in: USGS Professional Paper 856, 12 pp.

Flinders, A.F., Ito, G. & Garcia, M.O., 2010. Gravity anomalies of the Northern Hawaiian Islands: implications on the shield evolutions of Kauai and Niihau. *J. geophys. Res.,* **115,** doi: 10.1029/2009JB006877.

Flinders, A.F., Ito, G., Garcia, M.O., Sinton, J.M., Kauahikaua, J. & Taylor, B., 2013. Intrusive dike complexes, cumulate cores, and the extrusive growth of Hawaiian volcanoes. *Geophys. Res. Lett.,* **40,** 3367–3373.

Ge, S., Lin, G., Amelung, F., Okubo, P.G., Swanson, D.A. & Yunjun, Z., 2019. The accommodation of the South Flank's motion by the Koa'e Fault system, Kīlauea, Hawai'i: insights from the June 2012 earthquake sequence. *J. geophys. Res.: Solid Earth,* **124,** 11116–11129.

Gillard, D., Rubin, A. & Okubo, P., 1996. Highly concentrated seismicity caused by deformation of Kilaueas deep magma system. *Nature,* **384,** 343–346.

Helz, R.T., 1987. Differentiation of Kilauea Iki lava lake, Kilauea volcano, Hawaii: an overview of past and current work. in: Mysen, B.O.(Ed.), *Magmatic Processes: Physiochemical Principles*. The Geochemical Society, p. 241–258.

Hildenbrand, T.G., Rosenbaum, J.G. & Kauahikaua, J.P., 1993. Aeromagnetic study of the Island of Hawaii. *J. geophys. Res.: Solid Earth,* **98,** 4099–4119.

Hoversten, G.M., Gasperikova, E., Mackie, R., Myer, D., Kauahikaua, J., Newman, G.A. & Cuevas, N., 2022. Magnetotelluric investigations of the Kīlauea Volcano, Hawaii. *J. geophys. Res.: Solid Earth,* **127,** doi: 10.1029/2022JB024418.

Hyndman, R.D. & Drury, M.J., 1976. The physical properties of oceanic basement rocks from deep drilling on the Mid-Atlantic Ridge. *J. geophys. Res.,* **81,** 4042–4052.

Kauahikaua, J.P., Hildenbrand, T. & Webring, M., 2000. Deep magmatic structures of Hawaiian volcanoes, imaged by three-dimensional gravity models. *Geology,* **28,** 883–886.

Keller, G.V., Gross, L.T., Murray, J.C. & Skokan, C., 1979. Results of an experimental drill Hole AT the summit of Kilauea Volcano, Hawaii. *J. Volc. Geotherm. Res.,* **5,** 345–385.

Keller, T. & Suckale, J., 2019. A continuum model of multi-phase reactive transport in igneous systems. *Geophys. J. Int.,* **219,** 185–222.

Klein, F.W., Koyanagi, R.Y., Nakata, J.S. & Tanigawa, W.R., 1987. The seismicity of Kilauea's magma system. in: Decker, R.W., Wright, T.L. & Stauffer, P.H.(Eds.), *Volcanism in Hawaii*. U.S. Government Printing Office, Washington D.C., pp. 1019–1186.

Lerner, A.H., Sublett, D.M., Wallace, P.J., Cauley, C. & Bodnar, R.J., 2024. Insights into magma storage depths and eruption controls at Kīlauea Volcano during explosive and effusive periods of the past 500 years based on melt and fluid inclusions. *Earth planet. Sci. Lett.,* **628,** 118579, doi: 10.1016/j.epsl.2024.118579.

Lin, G. & Okubo, P.G., 2016. A large refined catalog of earthquake relocations and focal mechanisms for the Island of Hawai'i and its seismotectonic implications. *J. geophys. Res.: Solid Earth,* **121,** 5031–5048.

Lin, G., Shearer, P.M., Matoza, R.S., Okubo, P.G. & Amelung, F., 2014. Three-dimensional seismic velocity structure of Mauna Loa and Kilauea volcanoes in Hawaii from local seismic tomography. *J. geophys. Res.: Solid Earth,* **119,** 4377–4392.

Lundgren, P. *et al.* 2013. Evolution of dike opening during the March 2011 Kamoamoa fissure eruption, Kīlauea Volcano, Hawai`i. *J. geophys. Res.: Solid Earth,* **118,** 897–914.

Macdonald, G.A., 1944. The 1840 eruption and crystal differentiation in the Kilauean magma column. *Am. J. Sci.,* **242,** 177–189.

Mandl, G., 1988. *Mechanics of Tectonic Faulting: Models and Basic Concepts*. Elsevier, Amsterdam.

Manghnani, M.H. & Woollard, G.P., 2013. Elastic wave velocities in Hawaiian rocks at pressures to ten kilobars, in *The Crust and Upper Mantle of the Pacific Area,* Volume **12,** pp. 501–516.

Montgomery-Brown, E.K., Poland, M.P. & Miklius, A., 2015. Delicate balance of magmatic-tectonic interaction at Kīlauea Volcano, Hawai'i, revealed from slow slip events, in *Hawaiian Volcanoes,* pp. by John Wiley & Sons, Inc. 269–288. Chapter 13 in Hawaiian Volcanoes: From Source to Surface, Geophysical Monograph 208. First EditionEdited by Rebecca Carey, Valérie Cayol, Michael Poland, and Dominique Weis.

Montgomery-Brown, E.K., Segall, P. & Miklius, A., 2009. Kilauea slow slip events: identification, source inversions, and relation to seismicity. *J. geophys. Res.,* **114,** doi: 10.1029/2008JB006074.

Montgomery-Brown, E.K., Sinnett, D.K., Larson, K.M., Poland, M.P., Segall, P. & Miklius, A., 2011. Spatiotemporal evolution of dike opening and décollement slip at Kīlauea Volcano, Hawai'i. *J. geophys. Res.,* **116,** doi: 10.1029/2010JB007762.

Montgomery-Brown, E.K., Sinnett, D.K., Poland, M., Segall, P., Orr, T., Zebker, H. & Miklius, A., 2010. Geodetic evidence for en echelon dike emplacement and concurrent slow slip during the June 2007 intrusion and eruption at Kīlauea volcano, Hawaii. *J. geophys. Res.,* **115,** doi: 10.1029/2009JB006658.

Mosbrucker, A.R., Zoeller, M.H. & Ramsey, D.W., 2020. *Digital Elevation Model of Kīlauea Volcano, Hawai'i, Based on July 2019 Airborne Lidar Surveys*. U.S. Geological Survey.

Neal, C.A. *et al.*, 2019. The 2018 rift eruption and summit collapse of Kīlauea Volcano. *Science,* **363,** 367–374.

Nicolas, A. & Boudier, F., 1991. Rooting of the sheeted dikecomplex in the Oman ophiolite. in: Peters, T., Nicolas, A. & Coleman, R.G.(Eds.), *Ophiolite Genesis and Evolution of the Oceanic Lithosphere*. Kluwer Acad., Dordrecht, Netherlands, pp. 39–54.

Owen, S., Segall, P., Lisowski, M., Miklius, A., Denlinger, R. & Sako, M., 2000. Rapid deformation of Kilauea Volcano: global Positioning System measurements between 1990 and 1996. *J. geophys. Res.: Solid Earth,* **105,** 18983–18998.

Poland, M.P., Miklius, A., Jeff Sutton, A. & Thornber, C.R., 2012. A mantle-driven surge in magma supply to Kīlauea Volcano during 2003–2007. *Nat. Geosci.,* **5,** 295–300.

Ryan, M.P., 1988. The mechanics and three-dimensional internal structure of active magmatic systems: kilauea Volcano, Hawaii. *J. geophys. Res.: Solid Earth,* **93,** 4213–4248.

Segall, P., Anderson, K.R., Johanson, I. & Miklius, A., 2019. Mechanics of inflationary deformation during Caldera collapse: evidence from the 2018 Kīlauea eruption. *Geophys. Res. Lett.,* **46,** 11782–11789.

Shelly, D.R. & Thelen, W.A., 2019. Anatomy of a Caldera collapse: kīlauea 2018 summit seismicity sequence in high resolution. *Geophys. Res. Lett.,* **46,** 14395–14403.

Suckale, J., Keller, T., Cashman, K.V. & Persson, P.O., 2016. Flow-to-fracture transition in a volcanic mush plug may govern normal eruptions at Stromboli. *Geophys. Res. Lett.,* **43**(23), 12,071–12,081.

Turner, I.M., Peirce, C. & Sinha, M.C., 1999. Seismic imaging of the axial region of the Valu Fa Ridge, Lau Basin—the accretionary processes of an intermediate back-arc spreading ridge. *Geophys. J. Int.,* **138,** 495–519.

Veloso, E.E., Hayman, N.W., Anma, R., Tominaga, M., González, R.T., Yamazaki, T. & Astudillo, N., 2014. Magma flow directions in the sheeted dike complex at superfast spreading mid-ocean ridges: insights from IODP Hole 1256D, Eastern Pacific. *Geochem. Geophys. Geosyst.,* **15,** 1283–1295.

Walker, G.P.L., 1986. Koolau Dike Complex, Oahu: intensity and origin of a sheeted-dike complex high in a Hawaiian volcanic edifice. *Geology,* **14,** 310–313.

Wang, T., Zheng, Y., Pulvirenti, F. & Segall, P., 2021. Post-2018 Caldera collapse Re-inflation uniquely constrains Kīlauea's magmatic system. *J. geophys. Res.: Solid Earth,* **126**(6), e2021JB02180, doi: 10.1029/2021JB021803.

Wyss, M., Liang, B., Tanigawa, W.R. & Wu, X., 2012. Comparison of orientations of stress and strain tensors based on fault plane solutions in Kaoiki, Hawaii. *J. geophys. Res.: Solid Earth,* **97,** 4769–4790.

Zucca, J.J. & Hill, D.P., 1980. Crustal structure of the southeast flank of Kilauea Volcano, Hawaii, from seismic refraction measurements. *Bull. seismol. Soc. Am.,* **70,** 1149–1159.

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